waiyuamandang@gmail.com
amanda-ng518.github.io/
amanda-ng518
waiyuamanda-ng

Wai Yu Amanda Ng

Education

Sep 2025 – Apr 2026 **MSc in Statistics**, *University of Toronto*, Toronto, ON, Canada **cGPA: 4.00/4.00**
Courses: Applied Multivariate Analysis, Methods of Applied Statistics, Special Topics in Data Science, Survival Analysis

Sep 2021 – Apr 2025 **H.B.Sc in Statistical Science (Specialist), Mathematics (Major), Health Studies (Focus)**, *University of Toronto*, Toronto, ON, Canada **cGPA: 3.94/4.00**
Courses: Applied Statistical Methods, Applied Spatial Analysis, Time Series Analysis, Applied Bayesian Statistics, Surveys, Sampling and Observational Data, Statistical Consultation
Awards: University of Toronto Excellence Award (2024), Eleventh Annual Canadian Statistics Student Conference Undergraduate Level First Prize (2023)

Highlight of Skills

- 4+ years conducting data analysis across education and healthcare domains, including advanced querying, validation, predictive modeling and machine learning.
- Skilled in identifying trends, risks, and patterns in high-volume data and translating analytical findings into clear, actionable recommendations for technical and non-technical stakeholders.
- Strong problem-solving skills, detail-oriented, with experience delivering high-quality analytical work under tight timelines in collaborative, cross-functional environments.
- Experience working within a Linux environment and HPC environment (supercomputer).
- Tools: SQL, R, Python (Pandas, NumPy, Matplotlib, PyMC, Scikit-learn, Plotly), Excel, Tableau, Power BI, Git Version Control, QGIS, Azure Databricks
- Languages: English, Cantonese, Mandarin

Professional Experiences

Sep 2025 – Apr 2026 **Statistical Learning Designer**, *University of Toronto Human Biology Program*
- Built 8 auto-testable R assignments for a cohort of 400 students, cutting manual grading workload by 90% and accelerating feedback turnaround.
- Designed interactive LearnR, LearnSQL and LearnPython websites, reducing student debugging requests and enhancing self-directed learning efficiency.
- Standardized version-controlled releases on Git, enabling multi-developer collaboration and achieving zero student-reported issues across all instructional materials.

Sep 2024 – Apr 2025 **Machine Learning Researcher**, *University of Toronto Department of Statistical Sciences*
- Developed R/Python ETL pipeline (dplyr, Pandas, PyTorch) on 40,000+ EHR records from MIMIC-III database, producing a clean, standardized dataset that improved downstream model stability and interpretability.
- Performed feature selection on 6,000+ medical concept embeddings, supporting early identification of depression-related risk factors and informing modelling decisions.
- Developed and evaluated predictive models (logistic regression, random forests) achieving 85%+ accuracy, with fairness analyses identifying demographic disparities and improving model interpretability.
- Created reproducible analytical workflow and documentation and curated a GitHub repository, enhancing transparency and supporting sustainable analytical practices for future projects.

Sep 2024 – Apr 2025 **Data Visualization Analyst**, *University of Toronto Human Biology Program*
- Analyzed engagement and assessment data for 1,000+ learners in Excel, identifying early performance indicators that strengthened curriculum evaluation and learner-progress monitoring.
- Developed 100+ Tableau dashboards visualizing key metrics (discussion activity, video completion, early- term quiz performance), enabling instructors to identify emerging at-risk learners by Week 3.
- Communicated analytical findings into data-driven recommendations that improved assessment design, improving assessment design and instructional planning, such as course-delivery pacing.

Sep 2024 – Apr 2025 **Educational Data Analyst**, *University of Toronto Office of the Vice-Provost, Innovations in Undergraduate Education*
- Cleaned 3,000+ student Quercus engagement records across different time windows and designed standardized access frequency measures, enabling clear identification of disengagement patterns across school term.
- Built and validated logistic regression models in R to assess the relationship between course activity levels and dropout risk, revealing the first four weeks as a critical intervention period and supporting resources capacity-planning decisions.
- Delivered stakeholder-focused reports and presentations with clear metrics, visualizations, and narrative insights, informing revisions to early-term outreach and academic support strategies.

Jun 2024 – Apr 2025 **Clinical Education Research Analyst**, *Centre for Addiction and Mental Health*
- Analyzed 90+ VR medical training evaluation surveys in R using descriptive and non-parametric statistical methods to identify movement-related predictors of cybersickness.
- Identified a positive movement–nausea relationship, revealing key factors (e.g., frequent turning) contributing to user discomfort in clinical training, supporting quality-improvement decisions in VR training module design.
- Collaborated with clinical educators to translate analytical findings into evidence-based movement thresholds, resulting in a first author peer-reviewed publication.

Sep 2023 – Apr 2024 **Agent-Based Modeling Researcher**, *University of Toronto COBWEB lab*
- Partnered with environmental researchers to design 20+ agent-based simulations modeling population dynamics under declining birth rates, identifying two policy levers that improved long-term population stability.
- Synthesized 30+ multidisciplinary sources into an annotated bibliography, strengthening model assumptions and improving simulation realism.
- Conducted time-series and regression analyses in Python (SciPy) to quantify intervention effects, revealing short- vs long-term trade-offs that informed policy recommendations, resulting in a poster presentation.

May 2023 – Aug 2023 **Data Analyst Intern**, *Canadian Urban Institute*
- Integrated multi-source nationwide demographic and carbon-emission datasets to produce 300+ maps in QGIS, revealing regional household carbon-emission patterns and supporting targeted interventions for sustainable city-planning initiatives.
- Built regression models in R to identify location-specific emission drivers, informing long-term policy and operational planning (urbanization and housing-mix strategies) across multiple municipalities.
- Developed a interactive scenario-modelling dashboard enabling stakeholders to simulate policy impacts and strategic options.

Oct 2022 – May 2023 **Computational Social Science Researcher**, *University of Toronto Population Well-being Lab*
- Applied unsupervised machine learning methods (Gaussian mixture model, k-means, hierarchical clustering) to 150K+ population health survey observations, uncovering well-being subpopulation patterns based on life satisfaction and demographics.
- Created accessible statistical concepts tip-sheets, strengthening methodological literacy across lab members.
- Collaborated with faculty and graduate researchers to design a poster on clustering workflows and result interpretation, earning first prize at Statistic Society of Canada conference.

Research/Course Projects

Oct 2025 – Dec 2025 **Uncovering Calendar-Based Structures Through Clustering of Daily Traffic Data**
- Applied PCA and UMAP in R to a year-long traffic dataset across 26 locations, uncovering recurring traffic pattern clusters tied to school calendars, holidays, and seasonal routines.
- Created visualizations and prepared a report informing transportation planning and infrastructure maintenance decisions.

Oct 2025 – Dec 2025 **Fatal US School Shootings (1999-2025)**
- Modeled fatality risk in 400+ US school shootings using logistic regression in R, highlighting actionable risk indicators including shooting intent, absence of non-shooter injuries, and shooter connection to the school community.
- Built a reproducible GitHub repository documenting all data cleaning, modeling, and analysis steps to support transparency and collaboration.
- Synthesized findings into a report informing safety and emergency preparedness strategies.

Sep 2025 – Dec 2025 **Statistical Inference for Conditional Spatialtemporal Correlation in Brain MRI Data**, *Graduate Research Project*
- Pioneered analysis framework for assessing genetic heritability of intermodal coupling using multi-modal brain MRI data, generating methodological insights for advanced neuroimaging inference.
- Conducted 50+ HPC simulation experiments and applied adaptive clustering techniques with parallel computing to validate proposed model's Type I error control and statistical power.
- Summarized findings in a report advancing robust methodological neuroimaging statistical approaches.

Feb 2025 – Apr 2025 **Modeling Cocoa Futures Prices: A Statistical Analysis of Ghana's Cocoa Production**
- Integrated 3,700+ daily observations across four datasets, performing extensive cleaning to produce a consistent, analysis-ready dataset, enabling reliable comparison across climate, economic, and price variables.
- Developed ARIMA and Random Forest forecasting models in R (astsa, forecast, randomForest), engineering lagged features and covariates that improved predictive accuracy by capturing both linear and nonlinear drivers of cocoa price volatility.
- Conducted full diagnostics and authored a report translating model findings into actionable insights, highlighting need for additional economic and production data in future forecasting systems.

Oct 2024 – Dec 2024 **Transit and Crime: Investigating Assault Clusters Around Toronto's TTC Subway Lines**
- Analyzed 200k+ Toronto assault incidents using kernel density estimation, point process models, and HDBSCAN clustering, identifying assault hotspots within 100–500m of TTC subway lines, especially around key transfer stations.
- Developed an interactive geospatial report translating complex spatial patterns into map-based insights, providing location recommendations for targeted patrol allocation and station lighting improvement.

Sep 2024 – Apr 2025 **Quercus Engagement as a Predictor of First-Year University Student Dropout Risk: A Logistic Regression Analysis**, *Statistical Consulting Course*
- Cleaned 3,000+ student Quercus engagement records across different time windows and designed standardized access frequency measures, enabling clear identification of disengagement patterns across school term.
- Built and validated logistic regression models in R to assess the relationship between course activity levels and dropout risk, revealing the first four weeks as a critical intervention period and supporting resources capacity-planning decisions.
- Delivered stakeholder-focused reports and presentations with clear metrics, visualizations, and narrative insights, informing revisions to early-term outreach and academic support strategies.

Sep 2024 – Apr 2025 **Fair Machine Learning in Healthcare**, *Undergraduate Reading Course*
- Developed R/Python data pipeline (dplyr, Pandas, PyTorch) on 40,000+ EHR records from MIMIC-III database, producing a clean, standardized dataset that improved downstream model stability and interpretability.
- Developed embeddings for 6,000+ medical concepts and applied feature selection algorithms, yielding clinically relevant features that drove over 85% accuracy in subsequent prediction tasks.
- Trained logistic regression and random forest models for depression diagnosis, identifying key demographic disparities, such as race and insurance type, linked to label bias through fairness analysis.
- Built a reproducible workflow and curated a GitHub repository, enhancing research transparency and replicability.

May 2024 – Apr 2026 **Aspect of Robust Regression Analysis**, *Undergraduate Research Project*
- Developed and evaluated 4 parameter-estimation approaches for high-dimensional, heavy-tailed data, producing methodological insights that improved estimation accuracy over traditional regression methods.
- Implemented and optimized algorithms in R/Python, resolving non-convergence issues and increasing computational efficiency for large-scale simulations.
- Delivered a departmental research talk and released an R package with 20+ reusable functions, strengthening reproducibility and statistical communication.

Feb 2024 – Apr 2024 **Investigating The Factors Affecting Cycling Accident Severity: A Bayesian Hierarchical Mixture Model Approach**
- Conducted EDA on 800K+ accident records in Python (Pandas, NumPy, Matplotlib, Seaborn), identifying key temporal and environmental predictors, such as evening and dim environment, of severe injuries.
- Restoring 60K+ incomplete records by engineering Bayesian missing-data imputation pipeline (Scipy) using posterior predictive sampling, reducing model bias by 18% in downstream severity estimates.
- Built a Bayesian hierarchical mixture model in PyMC incorporating time-of-day structure, spike-and-slab selection, and demographic mixture components, achieving predictive accuracy gains over baseline logistic regression.

Sep 2023 – Apr 2024 **An Agent-Based Simulation Approach to Investigate the Effect of Decreasing Birth Rates and the Efficacy of Potential Solutions**, *Undergraduate Research Opportunity Program*
- Partnered with environmental researchers to design 20+ agent-based simulations modeling population dynamics under declining birth rates, identifying two policy levers that improved long-term population stability.
- Synthesized 30+ multidisciplinary sources into an annotated bibliography, strengthening model assumptions and improving simulation realism.
- Conducted time-series and regression analyses in Python (SciPy) to quantify intervention effects, revealing short- vs long-term trade-offs that informed policy recommendations, resulting in a poster presentation.

Health Studies Course Projects

Feb 2024 – Apr 2024 **Effectiveness of Usage of Virtual Reality in Raising Awareness about Cannabis Misuse Issues among Teenages**, *HMB342*
- Designed a randomized controlled trial study comparing virtual reality–based education with traditional seminars to evaluate effectiveness in raising awareness of cannabis misuse among Canadian high school students.
- Developed evaluation metrics and statistical analysis plan (pre/post surveys, T-tests, knowledge retention, behavioral intentions) to measure intervention impact on awareness and future cannabis use.
- Applied health promotion and program design frameworks to integrate emerging technologies into adolescent substance misuse education, highlighting cost-effectiveness and policy implications for public health strategies.

Feb 2023 – Apr 2023 **Food Insecurity Intervention Program Planning**, *HST330*
- Conducted root cause analysis and developed a program proposal to address food insecurity among Toronto low-income families, modeling intervention design on the Supplemental Nutrition Assistance Program with a digital e-coupon system.
- Designed a quasi-experimental evaluation plan, including baseline and post-intervention measures (Food Insecurity Experience Scale, BMI, Healthy Eating Index, stress levels) to assess program effectiveness.
- Created a logic model and theory of change framework to outline expected outcomes, health implications, and long-term policy applications, demonstrating strong skills in program evaluation and health policy analysis.

Oct 2022 – Dec 2022 **Proposal to Address Teenager Substance Misuse Problem**, *HST209*
- Conducted comprehensive analysis of risk factors (family influence, peer pressure, media exposure, mental health) contributing to teenage substance misuse, synthesizing evidence-based research into actionable insights.
- Designed and evaluated multi-level interventions including individual counseling, family therapy, in-school workshops, and peer-led support communities to promote adolescent health and reduce substance misuse.
- Developed health promotion strategies grounded in empowerment and community engagement, aligning interventions with public health models to improve long-term outcomes for at-risk youth.

Extracurricular

Sep 2023 – Apr 2026 **Teaching Assistant and Learning Support**, *University of Toronto Department of Statistical Sciences and Data Sciences Institute*
- Led tutorials and office hours across multiple statistics courses in person and online, strengthening undergraduate students and industry professionals understanding of R, Python, SQL, and statistical concepts.
- Supported high-school outreach events, Florence Nightingale Day, translating advanced statistical ideas into engaging, accessible explanations.

Sep 2022 – Dec 2023 **Statistical Consultant**, *University of Toronto Data Sciences Café*
- Provided on-demand statistical guidance to 20+ interdisciplinary researchers, transforming broad research questions into statistically rigorous analysis plans.
- Diagnosed design limitations and recommended tailored analytical approaches, helping researchers avoid common biases and strengthen study validity.
- Authored 5 reports on advanced statistical topics, translating complex concepts and coding methods into clear guidance that enhanced statistical literacy among non-statistician researchers.

Sep 2022 – Apr 2023 **First-year Learning Community Assistant Peer Mentor**, *University of Toronto*
- Facilitated 12 sessions supporting 20+ first-year students in academic planning, study strategies, and transition management.
- Built an inclusive learning community and coordinated discussions with faculty and library staff to enhance student engagement.
- Managed the online Quercus course page to facilitate remote communication and resource access.

May 2022 – Sep 2022 **Bilingual Client Care Representative**, *Telus Health*
- Served as first point of contact for 100+ daily client inquiries, delivering bilingual (English/Chinese) support that improved satisfaction and reduced escalations.
- Assessed client needs, identified the purpose of each inquiry, and provided tailored service recommendations, ensuring clients felt informed, supported, and guided toward the right solutions.
- Coordinated service delivery by booking appointments and routing callers to the appropriate service-specific teams, contributing to faster resolution times and smoother operations.